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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Based DevSecOps Secret Exposure Detection and Intelligent Security Risk Prioritization System	
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Abstract

AI-Based DevSecOps Secret Exposure Detection and Intelligent Security Risk Prioritization System is a security-focused solution designed to automatically detect sensitive information such as API keys, passwords, access tokens, database credentials, and cloud secrets exposed within source code and configuration files. The system integrates secret scanning into the DevSecOps pipeline so that security issues can be identified early during software development rather than after deployment.

The proposed system uses pattern-based detection, entropy analysis, and AI/ML-based classification to identify potential secrets while reducing false positives. Each detected vulnerability is analyzed based on factors such as secret type, exposure level, repository context, severity, and potential impact. An intelligent risk-prioritization module then assigns a risk score and categorizes vulnerabilities according to their urgency, helping developers address the most critical security issues first.

The system can be integrated with Git/GitHub, CI/CD pipelines, and cloud development environments to provide automated security checks during code commits and builds. It generates security reports containing detected secrets, risk levels, affected files, remediation recommendations, and security trends. By combining AI-driven analysis with DevSecOps automation, the proposed system aims to improve secure coding practices, reduce secret leakage, accelerate vulnerability remediation, and strengthen the overall security of modern software development workflows.

Signature of the Guide HOD